

Notes: Abis and Veldkamp (RFS, 2024)

The Changing Economics of Knowledge Production

Contents

1 Big picture

2 Data and measurement (key objects)

- 2.1 Why investment management is a useful laboratory
- 2.2 Main data sources
- 2.3 Job categories
- 2.4 Knowledge production functions
- 2.5 Structured data accumulation
- 2.6 Labor stocks from job postings
- 2.7 Initial data stock and depreciation
- 2.8 Wages and compensation
- 2.9 Descriptive facts

3 Models and empirical specifications (all equations + interpretation)

- 3.1 The two-layer production structure
- 3.2 Firm problem and value function
- 3.3 Labor first-order conditions
- 3.4 Data-management first-order condition
- 3.5 Optimal data stock condition
- 3.6 Iterative estimation with firm and time productivity effects
- 3.7 Why data-labor ratios are so revealing
- 3.8 Value of data

4 Identification and instruments (what makes the estimates credible)

- 4.1 There is no external IV; identification is structural

4.2	What primarily identifies α and γ	
4.3	Why the focus on investment management matters for identification	
4.4	Why wages can be interpreted as marginal-value objects	
4.5	Omitted inputs and the role of IT capital	
4.6	Depreciation and initial-condition robustness	
4.7	What the paper can and cannot distinguish	
5	Tables and figures	
5.1	Figure 1: production process for knowledge	
5.2	Figure 2: job classification from full text	
5.3	Figure 3 and Table 1: labor stocks and descriptive scale	
5.4	Figure 4: AI jobs pay more	
5.5	Table 2: the main production-function result	
5.6	Figure 5: the simple identifying statistic	
5.7	Figure 6: data value and analyst productivity	
6	Short summary	
7	Conclusion	
7.1	Core conclusion (what the paper establishes)	
7.2	Economic intuition	
7.3	Takeaway	

1 Big picture

- **Question.** Does big data and AI represent a modest improvement in statistical analysis, or a deeper technological change in the way firms produce knowledge?
- **Setting.** The paper studies the investment management industry using U.S. job-posting data from Burning Glass, salary data from Payscale, and BLS vacancy-fill and separation data. The structural estimation is done on 2015–2018, with 2010–2014 used as a burn-in period for labor stocks.
- **Core challenge.** Firms do not report their stock of data, their stock of knowledge, or the exact production technology they use. The paper has to infer all of these from observable hiring, skills, and wages.
- **Main contribution.** The paper builds a two-layer production model in which data managers transform raw information into structured data, and analysts combine structured data with labor to produce knowledge. This lets the authors estimate knowledge production functions for old technology and AI-based analysis.
- **Main hypothesis.** If AI is the industrialization of knowledge production, it should reduce the diminishing returns to data. In the model, that means the exponent on data should be larger under AI than under traditional analysis: $\alpha > \gamma$.
- **Main result.** In the baseline specification with 3% monthly data depreciation, the estimated exponent on data rises from $\gamma = 0.821$ for old-technology analysis to $\alpha = 0.868$ for AI analysis, a difference of about 0.05.
- **Income-share implication.** Because labor's share inside each Cobb-Douglas technology is $1 - \alpha$ or $1 - \gamma$, the shift from old technology to AI implies about a 4.7% decline in the share of knowledge income paid to labor and a corresponding rise in the share paid to data owners.
- **Magnitude.** The authors compare this 5% shift in factor shares to historical estimates from the industrial revolution and argue that it is comparable to the lower end of those estimates.
- **Labor-demand result.** The paper does *not* find that AI crowds workers out in finance. AI analyst jobs rise dramatically, but old-technology analyst jobs also continue to grow, so total analysis labor expands even though production becomes less labor-intensive.
- **Data-valuation result.** The estimated aggregate value of data for firms in the sample rises from about \$660 billion to about \$810 billion between 2015 and 2018, a 23% increase.
- **Broader takeaway.** New data technologies appear to make knowledge production more data-intensive, raise the value of data as an asset, reduce labor's share within knowledge production, and plausibly improve financial-market efficiency and real investment.

2 Data and measurement (key objects)

2.1 Why investment management is a useful laboratory

- The paper's model is about knowledge production in general, but the empirical work focuses on investment management because the sector is an early adopter of AI and big-data tools.

- Investment management is almost a pure knowledge industry: firms process information to form forecasts, signals, and trading strategies rather than making physical goods.
- Finance also has unusually high vacancy-fill rates. That matters because the paper observes job postings, not workers directly. If finance firms usually fill the jobs they post, postings are informative about desired labor input.
- Restricting attention to one industry also improves job classification, because job titles and skill language vary a lot across sectors.

Interpretation. The authors choose an industry where the measurement problem is tractable, the adoption of AI is already visible, and job postings are more likely to map into actual labor demand.

2.2 Main data sources

- **Burning Glass** job postings, January 2010 to December 2018. These postings come from more than 40,000 sources and include employer name, job title, skill requirements, and full job text.
- **Payscale** salary profiles. These provide total compensation, including salary, bonus, and profit share, for relevant occupations.
- **BLS** vacancy-fill and separation rates, used to turn job postings into labor-flow estimates.
- Treasury-bill rates and a sectoral risk premium, used to construct the discount rate in the value function.

Interpretation. The paper’s core idea is to combine quantity information from the hiring side with price information from wages. Hiring tells the authors about labor and data accumulation; wages tell them about factor income shares.

2.3 Job categories

The paper classifies jobs into three mutually distinct groups:

- **AI analysts** (L_{it}^{AI}): analysts using machine-learning or big-data tools.
- **Old-technology analysts** (L_{it}^{OT}): analysts using more traditional statistical and financial-analysis methods.
- **Data managers** (L_{it}^{DM}): workers who clean, structure, warehouse, integrate, or otherwise prepare data for analysis.

The classification uses the full text of job postings, not just titles. Figure 2 in the paper shows that the word frequencies are sharply different across these three groups.

Interpretation. The classification is crucial. The whole exercise depends on separating workers who *create structured data* from workers who *analyze* it, and also on separating AI-based analysis from older forms of analysis.

2.4 Knowledge production functions

The model assumes two knowledge-production technologies:

$$K_{it}^{AI} = A_t^{AI} a_i^{AI} D_{it}^\alpha (L_{it}^{AI})^{1-\alpha}, \quad (1)$$

$$K_{it}^{OT} = A_t^{OT} a_i^{OT} D_{it}^\gamma (L_{it}^{OT})^{1-\gamma}. \quad (2)$$

Here:

- D_{it} is the firm's stock of structured data,
- A_t^{AI} and A_t^{OT} are time-varying productivities,
- a_i^{AI} and a_i^{OT} are firm-specific productivities,
- α and γ govern the importance of data relative to labor.

Interpretation. A larger exponent on data means weaker diminishing returns to data and, equivalently, a smaller labor share inside that production technology. So the comparison $\alpha - \gamma$ is the paper's core measure of technological change.

2.5 Structured data accumulation

Structured data is produced by data-management labor according to

$$D_{i,t+1} = (1 - \delta)D_{it} + (L_{it}^{DM})^{1-\phi}. \quad (3)$$

Unfolding the recursion gives

$$D_{i,t+1} = D_{i0}(1 - \delta)^{t+1} + \sum_{s=0}^t (1 - \delta)^{t-s} (L_{is}^{DM})^{1-\phi}. \quad (4)$$

Interpretation. Data does not arrive for free. The firm must hire data managers to keep building and maintaining a searchable, usable data stock. The parameter ϕ governs diminishing returns in data creation, and δ governs depreciation.

2.6 Labor stocks from job postings

Because postings are flows rather than stocks, the paper cumulates them using fill and separation rates. For each type $x \in \{AI, OT, DM\}$,

$$L_{it}^x = (1 - s_t^x)L_{i,t-1}^x + j_{it}^x h_t^x, \quad (5)$$

where:

- j_{it}^x is the number of posted vacancies,
- h_t^x is the fraction of posted jobs that get filled,
- s_t^x is the separation rate.

The authors initialize labor at zero in 2010 and use 2010–2014 as burn-in, then estimate the model on 2015–2018.

Interpretation. The burn-in is a practical way to deal with missing initial conditions. By the time the estimation window starts, the cumulated labor series has largely converged to a meaningful stock.

2.7 Initial data stock and depreciation

The stock of data also requires an initial condition. The paper assumes each firm’s initial data stock is proportional to its first observed data inflow and then chooses the proportionality constant to best rationalize subsequent hiring choices through the optimality condition for data.

The paper explores monthly depreciation rates of 1%, 3%, and 10%.

- 1% corresponds roughly to software or infrastructure-style depreciation.
- 3% matches common amortization of data warehouses over about 36 months.
- 10% is a high-depreciation benchmark.

Interpretation. Depreciation matters because it determines how much past data-management labor contributes to today’s data stock. The authors treat 3% as the medium baseline and show the main conclusion is robust.

2.8 Wages and compensation

Payscale gives the paper the price side of the problem. The authors identify 11,041 salary profiles relevant to the investment-management jobs they study:

- 5,639 data-management profiles,
- 2,817 old-technology analyst profiles,
- 2,585 AI-analyst profiles.

Every month, the paper computes the mean compensation in each category.

Figure 4 shows that AI jobs pay on average about \$26,333 more per year than old-technology or data-management jobs.

Interpretation. The wage gap is informative because it says the market values AI analysis labor more highly. Since the production exponents are backed out from wage shares, good wage measurement is central to the whole exercise.

2.9 Descriptive facts

The labor-stock summary statistics in Table 1 show large cross-firm dispersion:

- mean data-management labor stock: 53.73,
- mean AI analyst labor stock: 2.33,

- mean traditional analyst labor stock: 32.37.

The dynamic patterns are also important:

- from 2015 to 2018, the stock of AI analysts rises from about 350 to about 4,537, roughly a 13-fold increase,
- by the end of 2018, around one-seventh of all financial analysts in the sample have AI-related skills,
- old-technology analyst employment also keeps growing.

Interpretation. AI labor is still small relative to total analysis labor, but it is growing fast. This is exactly the kind of transitional moment where one might hope to detect a shift in production technology.

3 Models and empirical specifications (all equations + interpretation)

3.1 The two-layer production structure

Figure 1 in the paper lays out the basic sequence:

raw/unstructured data $\rightarrow$ processing $\rightarrow$ structured data $\rightarrow$ analysis $\rightarrow$ knowledge.

The key point is that analysis does *not* work on raw data directly. Data must first be cleaned, structured, and made machine-readable by data managers.

Interpretation. This separation between data management and analysis is what allows the paper to value data as an accumulated productive asset rather than as an invisible by-product of labor.

3.2 Firm problem and value function

The representative firm solves

$$v(D_{it}) = \max_{L_{it}^{DM}, L_{it}^{AI}, L_{it}^{OT}} A_t^{AI} a_i^{AI} D_{it}^\alpha (L_{it}^{AI})^{1-\alpha} + A_t^{OT} a_i^{OT} D_{it}^\gamma (L_{it}^{OT})^{1-\gamma} - w_t^{AI} L_{it}^{AI} - w_t^{OT} L_{it}^{OT} - w_t^{DM} L_{it}^{DM} + \frac{1}{r} v(D_{i,t+1}), \quad (6)$$

subject to

$$D_{i,t+1} = (1 - \delta) D_{it} + (L_{it}^{DM})^{1-\phi}. \quad (7)$$

Interpretation. The firm chooses three types of labor today while recognizing that data is a state variable that carries value into the future.

3.3 Labor first-order conditions

For AI labor:

$$(1 - \alpha) K_{it}^{AI} - w_t^{AI} L_{it}^{AI} = 0. \quad (8)$$

For old-technology labor:

$$(1 - \gamma)K_{it}^{OT} - w_t^{OT}L_{it}^{OT} = 0. \quad (9)$$

Taking the ratio gives

$$\frac{(1 - \alpha)K_{it}^{AI}}{(1 - \gamma)K_{it}^{OT}} = \frac{w_t^{AI}L_{it}^{AI}}{w_t^{OT}L_{it}^{OT}}. \quad (10)$$

Interpretation. These conditions say that total wage payments equal the labor share of knowledge output. This is the key link between observed wages and the unobserved technology parameters.

3.4 Data-management first-order condition

The first-order condition for data-management labor is

$$\frac{1}{r}v'(D_{i,t+1})(1 - \phi)(L_{it}^{DM})^{-\phi} = w_t^{DM}. \quad (11)$$

Approximating the marginal value of data as sufficiently stable from one period to the next yields

$$\frac{(\alpha K_{it}^{AI} + \gamma K_{it}^{OT})(1 - \phi)}{r - (1 - \delta)} \frac{D_{i,t+1} - (1 - \delta)D_{it}}{D_{it}} - w_t^{DM}L_{it}^{DM} = 0. \quad (12)$$

Interpretation. Data-management pay reflects how much the firm's data stock grows and how much that data stock contributes to knowledge production.

3.5 Optimal data stock condition

Substituting the labor first-order conditions into the data-manager condition gives an operational equation for the firm's optimal data stock:

$$D_{it} - \left(\frac{\alpha}{1 - \alpha} w_t^{AI} L_{it}^{AI} + \frac{\gamma}{1 - \gamma} w_t^{OT} L_{it}^{OT} \right) \frac{(1 - \phi)(L_{it}^{DM})^{-\phi}}{(r - (1 - \delta))w_t^{DM}} = 0. \quad (13)$$

Interpretation. This equation is extremely important empirically: it says the optimal data stock can be inferred from observable hiring choices and wages. In the paper's words, it is the amount of data that best rationalizes the firm's labor decisions.

3.6 Iterative estimation with firm and time productivity effects

The structural estimation jointly recovers four unknowns:

$$(\alpha, \gamma, \phi, \bar{D}_0). \quad (14)$$

Given a guess for these parameters, the authors back out time-varying productivity terms from cross-firm averages of the first-order conditions,

$$A_t^{AI} = \frac{\omega_t^L \sum_i L_{it}^{AI}}{(1 - \alpha) \sum_i a_i^{AI} D_{it}^\alpha (L_{it}^{AI})^{1-\alpha}}, \quad A_t^{OT} = \frac{\omega_t^I \sum_i L_{it}^{OT}}{(1 - \gamma) \sum_i a_i^{OT} D_{it}^\gamma (L_{it}^{OT})^{1-\gamma}}, \quad (15)$$

and then recover firm-specific productivities from time averages,

$$a_i^{AI} = \frac{\sum_t \omega_t^L L_{it}^{AI}}{(1 - \alpha) \sum_t A_t^{AI} D_{it}^\alpha (L_{it}^{AI})^{1-\alpha}}, \quad a_i^{OT} = \frac{\sum_t \omega_t^I L_{it}^{OT}}{(1 - \gamma) \sum_t A_t^{OT} D_{it}^\gamma (L_{it}^{OT})^{1-\gamma}}. \quad (16)$$

They then minimize the stacked errors from the AI labor condition, the old-technology labor condition, the data-management labor condition, and the optimal data-stock condition.

Interpretation. The iterative procedure effectively absorbs firm fixed effects and time fixed effects. After this, the remaining firm-time variation is what identifies the technology parameters.

3.7 Why data-labor ratios are so revealing

A central empirical insight of the paper is that a more data-intensive technology should generate a stronger relationship between a firm's data stock and its payments to AI workers, relative to its payments to old-technology workers.

The paper writes this idea as

$$D_{it} = \beta_1 \frac{w_t^{OT} L_{it}^{OT}}{(L_{it}^{DM})^\phi w_t^{DM}} + \beta_2 \frac{w_t^{AI} L_{it}^{AI}}{(L_{it}^{DM})^\phi w_t^{DM}}, \quad (17)$$

with

$$\beta_1 = \frac{(1 - \phi)}{r - (1 - \delta)} \frac{\gamma}{1 - \gamma}, \quad \beta_2 = \frac{(1 - \phi)}{r - (1 - \delta)} \frac{\alpha}{1 - \alpha}. \quad (18)$$

Estimated slopes are about 64.92 for the OT/DM term and 76.83 for the AI/DM term, which implies that the AI slope is steeper and therefore that $\alpha > \gamma$.

Interpretation. This is the clearest reduced-form intuition behind the whole paper: firms that use more AI labor also hold disproportionately more data, even after scaling by data-management payments. That is exactly what the structural model predicts if AI lowers diminishing returns to data.

3.8 Value of data

Once the parameters and data stocks are estimated, the value function directly delivers a private value of data for each firm-month. Aggregating across firms, the paper estimates that data value rises from about \$660 billion in 2015 to about \$810 billion in 2018.

The paper also makes a rough economy-wide calculation. It maps additional data value into improved forecast precision, then into a lower cost of capital, and then compares that magnitude to the real effects of the EDGAR rollout. The implied effect is conservative but still economically meaningful: roughly a 5% increase in real investment.

Interpretation. The data-valuation exercise is one of the paper's most interesting by-products. The model is not only about factor shares; it is also a method for attaching a dollar value to data stocks.

4 Identification and instruments (what makes the estimates credible)

4.1 There is no external IV; identification is structural

This paper does not use an instrument in the usual reduced-form sense. Instead, identification comes from the combination of:

- the model’s first-order conditions,
- the distinction between AI labor, OT labor, and data-management labor,
- cross-firm and within-firm-time variation in labor stocks and wages,
- and the iterative removal of firm-specific and time-specific productivity terms.

Interpretation. Credibility therefore depends less on a single exclusion restriction and more on whether the observed hiring and wage patterns line up with the structural logic of the model.

4.2 What primarily identifies α and γ

The paper is explicit that no single data feature fully identifies the exponents, but one feature does most of the work: the sensitivity of a firm’s data stock to AI versus OT worker payments.

If AI truly makes data more productive, then firms using AI should optimally hold more data per unit of analysis labor. Figure 5 shows exactly that pattern.

Interpretation. The key identifying comparison is not merely “firms with AI pay higher wages.” It is “firms with AI labor choose higher data-labor ratios.”

4.3 Why the focus on investment management matters for identification

The empirical strategy requires very clean job classification and a labor market where posted jobs are likely to be filled. Investment management helps on both counts:

- the sector has industry-specific vocabulary that helps distinguish job types,
- the sector is an early adopter of AI,
- and finance vacancies tend to fill at relatively high rates.

Interpretation. The paper’s industry choice is part of the identification strategy, not just a convenience sample.

4.4 Why wages can be interpreted as marginal-value objects

A natural concern is that observed wages need not equal marginal products. The paper argues this problem is limited here because finance is a liquid labor market with high pay and high fill rates.

It would be irrational for firms to systematically pay above marginal product, and while frictions could make wages slightly lower than marginal value, the basic income-share interpretation should still be informative.

Interpretation. The wage assumption is strong, but the paper makes it in a setting where it is more plausible than in many other labor markets.

4.5 Omitted inputs and the role of IT capital

The knowledge production function omits capital, land, management quality, and other inputs. The authors argue that the omission is less problematic here than in standard industrial settings because knowledge production is an intermediate activity that may use relatively little physical capital. They also report appendix evidence that IT spending behaves more like a firm fixed effect than a margin that sharply rises with AI hiring.

Interpretation. If IT capital simply tracks persistent firm differences, the fixed-effect structure absorbs much of the concern. The paper does not claim capital is irrelevant; it claims the omission does not seem to overturn the estimated change in the data exponent.

4.6 Depreciation and initial-condition robustness

A major concern in any perpetual-inventory style exercise is that the results may be driven by arbitrary assumptions about the initial stock or the depreciation rate. The paper addresses this directly:

- it reports results for 1%, 3%, and 10% monthly depreciation,
- it treats 3% as the medium baseline,
- and it shows that many alternative specifications imply similar or even larger changes in the AI exponent.

Interpretation. The 5% baseline estimate is meant to be conservative, not cherry-picked.

4.7 What the paper can and cannot distinguish

The paper is careful about its boundaries.

- It identifies a change in the way labor and data combine to produce knowledge in one sector.
- It does *not* claim to measure the final-good production function or the full long-run general-equilibrium effect of AI.
- It does *not* pin down whether the relevant economic object is “data” itself or the broader automation system that continuously updates and deploys data.
- It studies a transition path, not the final steady state of the AI economy.

Interpretation. The results are strongest as an estimate of how AI changes knowledge production in investment management during an early adoption phase. Extrapolating to all sectors should be done carefully.

5 Tables and figures

5.1 Figure 1: production process for knowledge

Read:

- Figure 1 lays out the full production chain: raw data become structured data through processing, and structured data then combine with analyst labor to produce knowledge.
- The figure also makes visually clear that AI and old technology share the same data stock but use different labor inputs and different data exponents.

Economic message. This figure is the conceptual backbone of the paper. The whole estimation strategy depends on treating data management and data analysis as separate stages.

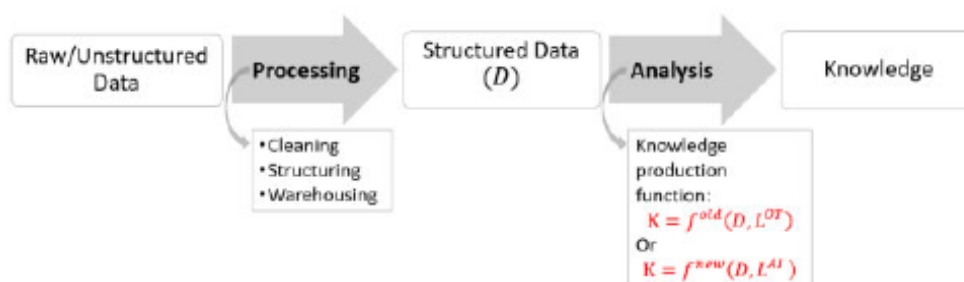


Figure 1
Production process for knowledge

The old technology knowledge production function is:

Figure 1 from Abis and Veldkamp (2024).

5.2 Figure 2: job classification from full text

Read:

- The AI word cloud emphasizes terms like machine learning, data science, and related technical tools.
- The old-technology cloud emphasizes statistics, math, portfolio management, and more traditional analysis language.
- The data-management cloud emphasizes databases, data models, SQL, and data quality.

Economic message. The paper's classification strategy is credible only if these occupations really look different in the text. Figure 2 shows that they do.

and categorize them.

What is important for our estimates is not that we count every job, or even every analyst job, in a firm. Because we use ratios of various types of jobs to identify our parameters of interest, what is important is that we have a representative sample of the jobs in a firm.

Figure 2 illustrates the frequency of all keywords in the samples of job postings categorized as AI, old tech, or data management (left to right subfigures), where a greater size indicates a higher frequency. Even if all



Figure 2 from Abis and Veldkamp (2024).

5.3 Figure 3 and Table 1: labor stocks and descriptive scale

Read:

- Figure 3 shows that the fraction of employers hiring AI labor rises sharply after 2015.
- The AI labor stock jumps from about 350 to about 4,537 between 2015 and 2018.
- Data-management and traditional-analysis labor stocks are much larger in levels, but AI grows much faster.
- Table 1 shows the stock distribution is highly dispersed across firms. Mean labor stocks are 53.73 for data managers, 32.37 for traditional analysts, and 2.33 for AI analysts.

Economic message. These exhibits tell us two things. First, AI adoption is real and economically meaningful. Second, the cross-firm heterogeneity needed to identify the technology parameters is very large.

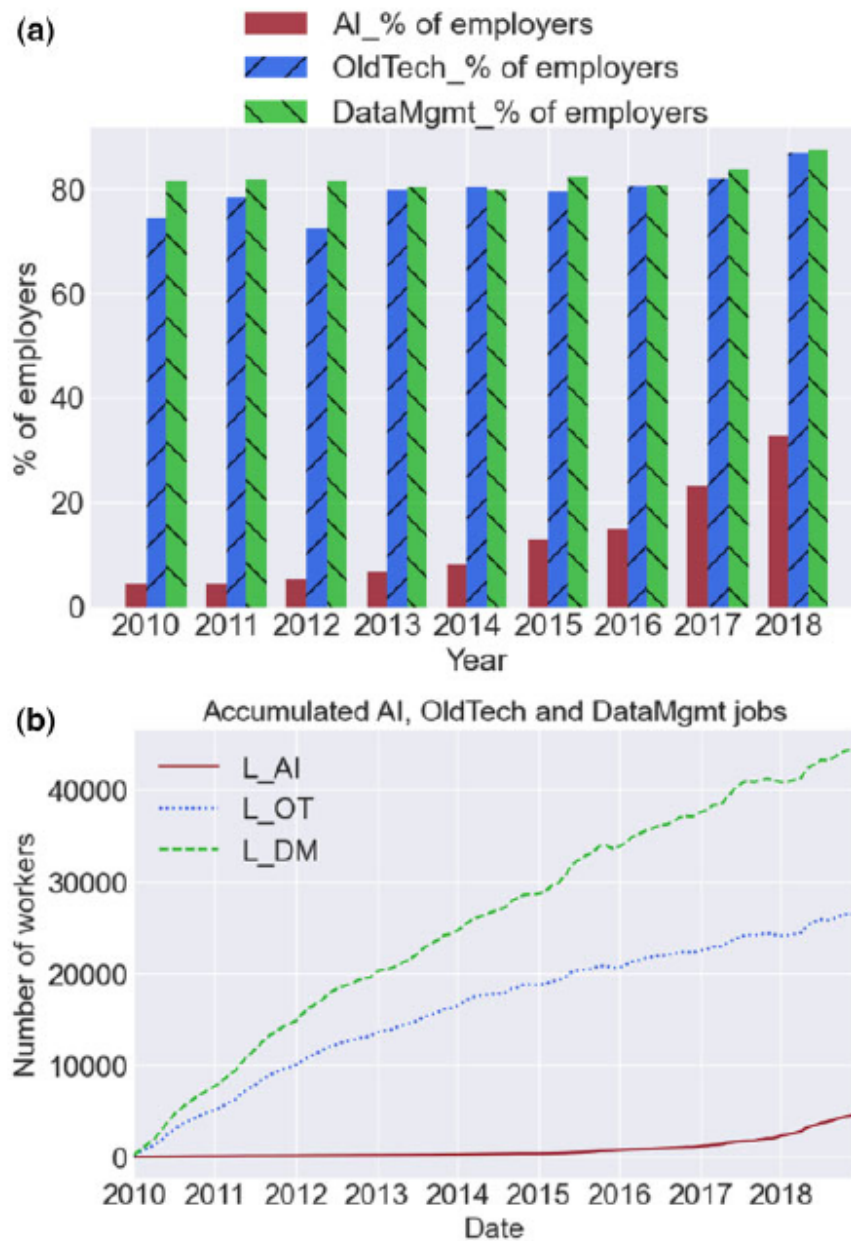


Figure 3

Job postings and labor stocks

Panel (a) shows the fraction of employers hiring in each category. Panel (b) shows the stock of labor in category, measured as a cumulated number of job postings, adjusting for filling and separation rates, as in Source: Burning Glass, 2010–2018.

Figure 3 from Abis and Veldkamp (2024).

Table 1
Labor stock summary statistics

	Data Management L_{it}^{DM}	AI analysts L_{it}^{AI}	Traditional analysts L_{it}^{OT}
mean	53.73	2.33	32.37
stdev	441.17	29.78	204.84
minimum	0	0	0
median	5.72	0	3.63
maximum	11,409.26	1,765.88	4,420.53
Observations	33,392	33,392	33,392

Distribution of the number of AI, old tech, or data management workers (stock of labor) per firm in a pooled sample of all firms and time periods of interest. Source: Burning Glass, 2015–2018.

Table 1 from Abis and Veldkamp (2024).

5.4 Figure 4: AI jobs pay more

Read:

- Figure 4 plots monthly average wages for AI analysts, old-technology analysts, and data managers.
- AI compensation is consistently above the other two categories.
- The average premium is about \$26,333 per year.

Economic message. The wage data support the idea that AI analysis labor is more productive and reinforce the interpretation of wages as informative about factor shares and marginal values.



Figure 4 from Abis and Veldkamp (2024).

5.5 Table 2: the main production-function result

Read:

- Under the baseline 3% monthly depreciation, the estimates are $\alpha = 0.868$, $\gamma = 0.821$, and $\phi = 0.443$.
- At 1% depreciation, the estimated change is even larger: $\alpha - \gamma = 0.173$.
- At 10% depreciation, the difference is still about 0.047.

Economic message. Table 2 is the headline table. It shows that AI meaningfully reduces the diminishing returns to data and therefore reduces labor's share within knowledge production.

Table 2 Main result: AI reduced the share of knowledge income paid to labor ($\alpha > \gamma$)				
		$\delta = 1\%$	$\delta = 3\%$	$\delta = 10\%$
AI Analysis	α	0.924 (0.0010)	0.868 (0.0015)	0.694 (0.0018)
Old Technology Analysis	γ	0.751 (0.0029)	0.821 (0.0020)	0.647 (0.0018)
Data Management	ϕ	0.337 (0.0061)	0.443 (0.0058)	0.010 (0.0074)
Change in Labor Share	$\gamma - \alpha$	-17.3%	-4.7%	-4.7%

Table 2 from Abis and Veldkamp (2024).

5.6 Figure 5: the simple identifying statistic

Read:

- Figure 5 plots the relationship between data stocks and two scaled payment ratios: AI/DM and OT/DM.
- The AI/DM slope is visibly steeper than the OT/DM slope.
- The estimated slopes are 76.83 for AI/DM and 64.92 for OT/DM.

Economic message. This is the most intuitive visual summary of the main result: additional AI worker payments are associated with a larger increase in data than additional OT worker payments, exactly as the model predicts when AI makes large data sets more productive.

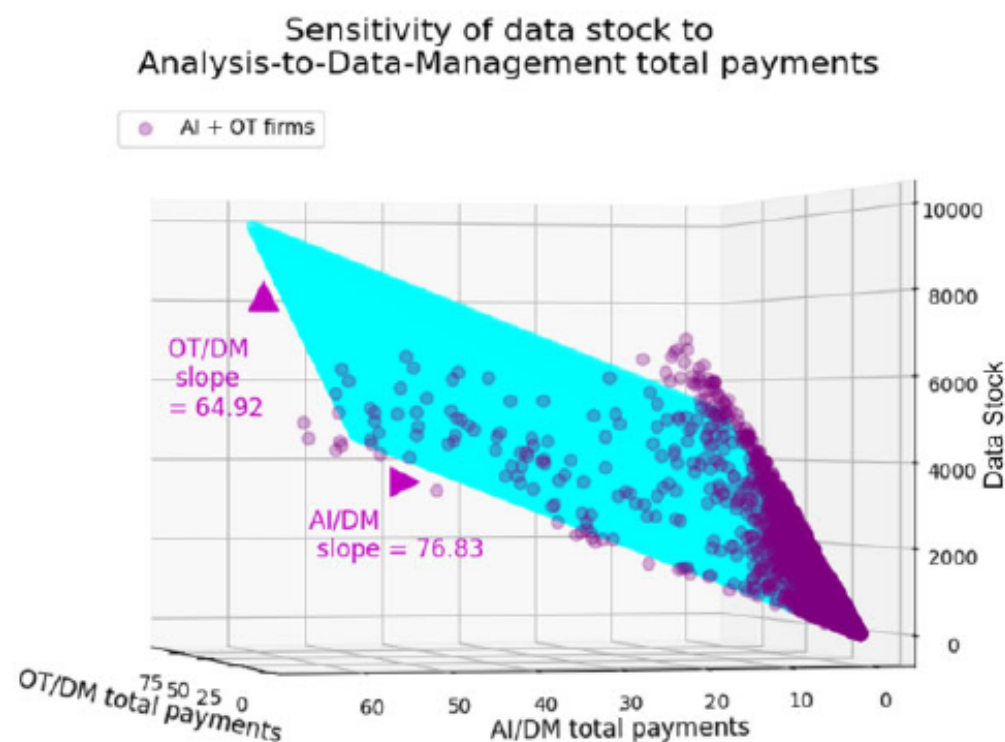


Figure 5 from Abis and Veldkamp (2024).

5.7 Figure 6: data value and analyst productivity

Read:

- Panel (a) shows the cumulative private value of data rising from roughly \$660 billion to roughly \$810 billion over 2015–2018.
- Panel (b) shows that AI productivity rises much faster than old-technology productivity.
- The paper attributes the increase in data value to more data accumulation, more analysts using data, and higher productivity in working with data.

Economic message. Figure 6 connects the production-function estimates back to economically interpretable objects. AI is not only changing exponents; it is increasing the dollar value of data and the productivity of knowledge production.

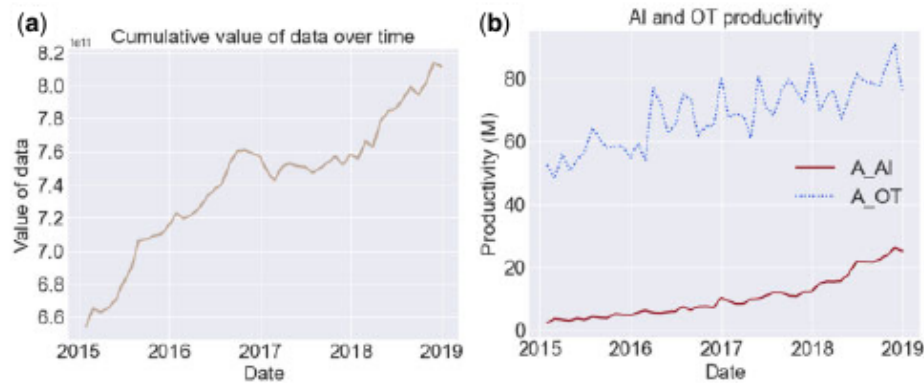


Figure 6 from Abis and Veldkamp (2024).

6 Short summary

- The paper studies whether AI changes the production function for knowledge.
- It models knowledge production with two analyst technologies – old technology and AI – and a separate data-management stage that builds the data stock.
- Using hiring and wage data from the investment management industry, it structurally estimates the exponent on data in both technologies.
- The central result is that AI raises the data exponent by about 0.05 in the baseline case, implying weaker diminishing returns to data and about a 4.7% decline in labor's share within knowledge production.
- The implied size of the shift is comparable to the lower end of estimates for the industrial revolution.
- There is no evidence that AI reduces total analyst employment in finance; instead, AI and old-technology labor both expand.
- The estimated private value of data rises from about \$660 billion to about \$810 billion over the sample period.
- The paper therefore portrays AI as a genuine factor-share-changing technology in knowledge production, not just a marginal improvement in office software.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that AI changes the way labor and data combine to produce knowledge. In the investment-management sector, AI makes knowledge production more data-intensive by reducing

the diminishing returns to data. In the baseline estimate, this shifts roughly 5% of knowledge income away from labor and toward data owners.

7.2 Economic intuition

Industrialization increased the importance of capital relative to labor in goods production. This paper argues that big-data technologies are doing something analogous in knowledge production. Once data can be accumulated, structured, and continuously updated, AI lets firms extract more value from large data stocks than older analytical methods could. That raises the return to data, not because labor disappears, but because each worker can now operate on a richer data environment.

7.3 Takeaway

The paper's main message is not simply that AI is useful. It is that AI is changing factor shares inside knowledge production, increasing the asset value of data, and potentially affecting the labor share of income and the efficiency of financial markets. In that sense, calling big data the “industrialization of knowledge production” is not just metaphorical; the paper shows that the comparison has real quantitative content.